

Algebra 1
Unit 5
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

1. Solve the equation and verify the solution using substitution.

$$\begin{aligned}
 10 \left[4 - \frac{1}{2}(2 + x) = \frac{2}{5}(-1 + 3x) \right] \\
 40 - 5(2 + x) &= 4(-1 + 3x) \\
 40 - 10 - 5x &= -4 + 12x \\
 30 - 5x &= -4 + 12x \\
 34 &= 17x \\
 x &= 2
 \end{aligned}$$

$$\begin{aligned}
 4 - \frac{1}{2}(2 + 2) &= \frac{2}{5}(-1 + 3 \cdot 2) \\
 4 - \frac{1}{2}(4) &= \frac{2}{5}(-1 + 6) \\
 4 - 2 &= \frac{2}{5}(5) \\
 2 &= 2 \checkmark
 \end{aligned}$$

The point-slope form of a line is given by the formula $y - y_1 = m(x - x_1)$.

2. Use the distributive property to rewrite the point-slope formula without parentheses.

$$y - y_1 = mx - mx_1$$

3. Use your answer from question 2 to rearrange the point-slope formula to solve for the variable y .

$$y - y_1 = mx - mx_1$$

$$\frac{+y_1}{+y_1} + y_1$$

$$y = mx - mx_1 + y_1$$

4. Compare your result in question 3 to the slope-intercept form of a line. Using your answer from question 3 what is the y -intercept in your equation?

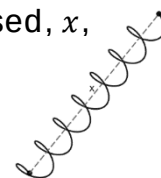
$$y_1$$

Rearrange the formulas to isolate the specified variables.

5. Hooke's Law, $F = -kx$, is used in physics to find the force, F , needed to stretch or compress a spring, based on the length the spring is stretched or compressed, x , and the spring constant, k . Solve the equation for x .

$$\frac{F}{-k} = \frac{-kx}{-k}$$

$$x = -\frac{F}{k}$$



6. The formula for the volume of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Solve the equation for the radius, r .

$$\frac{3}{4} \left[V = \frac{4}{3}\pi r^3 \right]$$

$$\frac{\frac{3}{4}V}{\pi} = \frac{\pi r^3}{\pi}$$

$$\sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{r^3}$$

$$r = \sqrt[3]{\frac{3V}{4\pi}}$$

7. The surface area of a rectangular prism is given by the formula $SA = 2(lw + lh + hw)$. Rearrange the formula to solve for the width, w .

$$SA = 2(lw + lh + hw)$$

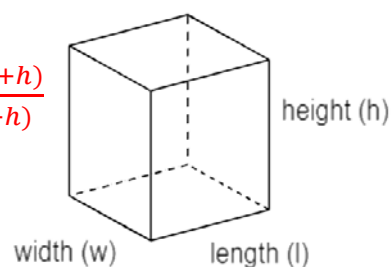
$$SA = 2lw + 2lh + 2hw$$

$$\frac{-2lh}{2(l+h)} \quad \frac{-2lh}{2(l+h)}$$

$$SA - 2lh = 2lw + 2hw$$

$$\frac{SA - 2lh}{2(l+h)} = \frac{2w(l+h)}{2(l+h)}$$

$$\frac{SA - 2lh}{2(l+h)} = w$$



The area of a composite shape can be found using several methods. One method is to divide the total area into sub-problems, then find the area of each one, and add them together to get the total area. Another method is to calculate the area of a larger, simpler figure and subtract off the unnecessary pieces. Daeun and Jose have been asked to find the area of the figure below. Daeun chooses to use sub-problems while Jose uses the subtraction method. Their work is shown below. Use to their work to answer the questions that follow.

Daeun's Work	Jose's Work
The area is equal to sum of the inside areas: - rectangle: $a \times b$ - rectangle: $a \times (d + e)$ - rectangle: $c \times d$ - triangle: $\frac{1}{2}e \times c$	The area is equal to the area of the large rectangle, minus the extra pieces: - large rectangle: $(a + c) \times (b + d + e)$ - extra rectangle: $b \times c$ - extra triangle: $\frac{1}{2}e \times c$
Total Area: $A = ab + cd + \frac{1}{2}ec + a(d + e)$	Total Area: $A = (a + c)(b + d + e) - bc - \frac{1}{2}ec$

8. Use the known lengths to label the lengths of the dashed lines (a, b, c, d, e , etc).
9. Simplify both equations for the total area so that there are no more parentheses in either one.

$$A = ab + cd + \frac{1}{2}ec + ad + ae$$

$$A = ab + ad + ae + cd + \frac{1}{2}ce$$

10. Did Daeun and Jose get the same answer for the total area? If so, rearrange Daeun's simplified equation so that it is the same as Jose's. If not, identify and correct any mistakes you find.

$$\text{Yes } A = ab + cd + \frac{1}{2}ec + ad + ae = ab + ad + ae + cd + \frac{1}{2}ce$$